

CHAPALA NAVEEN KUMAR

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

B.Tech — Computer Science (Artificial Intelligence & Machine Learning) 2024–2028
SIMATS Engineering College, Chennai CGPA: 8.8/10
Coursework: Machine Learning, Deep Learning, NLP, Artificial Intelligence, Probability & Statistics, Data Structures & Algorithms

Technical Skills

Programming: Python, SQL, Data Structures & Algorithms
Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Evaluation
Mathematics for ML: Linear Algebra, Probability, Statistics, Calculus
Deep Learning: CNN, ANN, RNN, LSTM, GRU, Transformers, TensorFlow, Keras
NLP & Generative AI: TF-IDF, Word Embeddings, Sentiment Analysis, LLM Integration, HuggingFace, LangChain, Langgraph, RAG, Prompt Engineering, Vector Databases
MLOps: Flask, FastAPI, Docker, MLflow, DVC, CI/CD, Model Deployment, Azure
Databases: DBMS, SQLite 3
Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Tools: Git, GitHub, VS Code, Jupyter Notebook, Streamlit, Linux

Projects

DataVision AI — Autonomous AutoML & Business Intelligence Platform [Live](#)
Tech Stack: Python, FastAPI, Scikit-learn, TensorFlow, NLTK, FAISS, LLMs, React, Docker, HuggingFace

- Developed production-grade AutoML platform enabling zero-code ML model training and analytics
- Implemented AutoML engine with 50+ ML algorithms, automated EDA and hyperparameter tuning
- Integrated LLM-powered AI analyst chatbot using RAG for business insights and forecasting
- Built scalable full-stack system with real-time dashboards and prediction APIs
- Improved model selection efficiency by 60% using automated hyperparameter optimization

Real-Time Stock Market Prediction using LSTM Deep Learning with Technical Indicators [Live](#)
Tech Stack: Python, TensorFlow, Keras, LSTM, Pandas, NumPy, Flask, React, HuggingFace, MLflow, DVC, CI/CD

- Built deep learning time-series forecasting model using LSTM networks
- Engineered technical indicators (RSI, MACD, SMA, EMA) to improve prediction performance
- Achieved **84% prediction accuracy** using optimized hyperparameters
- Deployed real-time stock forecasting web application with interactive dashboard

Diabetes Risk Prediction System [Live](#)
Tech Stack: Python, Scikit-learn, Pandas, NumPy, RandomForest, Flask, React, HuggingFace, MLflow, DVC, CI/CD

- Developed end-to-end machine learning pipeline for disease risk prediction
- Performed data preprocessing, feature engineering, and model optimization
- Achieved **87% accuracy** using ensemble machine learning models
- Deployed production-ready web application for real-time predictions

Certifications

Oracle OCI Generative AI Professional — Oracle (2025)
Machine Learning, Deep Learning & NLP Bootcamp — Udemy
Retrieval Augmented Generation (RAG) with LLMs — Udemy

Achievements

Best Student Award 2025 — Wisdom Path Foundation
Team Leader — Smart India Hackathon Participant
Built and deployed production-grade AutoML platform (DataVision AI)
Active participant in Machine Learning, AI, and hackathons